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Exploring the Link Between Mitochondrial RNA & Aging Through a Data-Driven Approach

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Abstract

Background: Mitochondrial function and gene expression have long been tightly linked to cellular aging and metabolic longevity. The exact nature of this relationship remains an open problem. However, empirical observations reveal a non-linear relationship, marked by elevated mitochondrial gene expression in early developmental and late-life stages alongside lower relative expression in healthy adult states.

Uncovering these subtle shifts requires specialized bioinformatic pipelines that isolate organellar transcripts (RNA molecules transcribed directly from non-nuclear organelle genomes, such as the mitochondria, which are profiled to quantify cellular energy production, metabolic state, and bioenergetic shifts across development, aging, and disease) while filtering out Nuclear Mitochondrial Pseudogenes (NUMTs are non-functional fragments of mitochondrial DNA integrated into the nuclear genome, which must be filtered out to avoid off-target alignment artifacts and false-positive variant calls when profiling true mitochondrial sequences) and sequencing noise.

This report evaluates a custom alignment pipeline using human CD4⁺ T-cell RNA sequencing data (NCBI SRA: SRP132689 / PRJNA433825) across age cohorts ranging from 21 weeks to 70 years old.

Methods: We benchmarked 16 FASTQ datasets (500,000 “single-end Illumina HiSeq 3000” reads per sample) aligned against the complete human mitochondrial genome reference (GenBank: EU007839). Donor age is denoted by w for gestational weeks (e.g., w_{21} indicates a 21-week-old fetal sample) and y for years (e.g., y_{26} indicates a 26-year-old adult).

We evaluated four distinct operational modes (k -mer identity match score filtering (`kmer`), dynamic Levenshtein edit distance filtering (`edit_distance`), dual-verification hybrid filtering (`both`), and an unfiltered control (`raw_unfiltered`)) across k -mer seed sizes ($k = 5, 7, 10$) and edit bounds ($e \leq 2, 3$). Code implementations are publicly available on GitHub.

Results: Pure k -mer filtering ($k = 5, 7, 10$) ($\text{Score} \geq 0.7$) captured up to **42,365** aligned reads (MT Frac ≈ 0.0847) and 94.05% genome coverage for sample `w21_SRR6706578_500k.fastq`, but permitted alignments with edit distances up to 15. Dual-verification hybrid filtering ($\text{Score} \geq 0.7$ and $e \leq 2$) successfully removed noisy nuclear artifacts, reducing mean edit distance from 0.13 to 0.10 and raising mean alignment scores to 0.9974. Across all configurations, captured mitochondrial read fractions ranged between 3.89% and 8.47% of total reads. This proportion aligns with standard biological expectations for cellular RNA-seq, validating pipeline precision. Furthermore, the pipeline identified a non-linear expression trajectory across age groups: high expression in fetal samples (w_{21}, w_{23} : MT Frac $\approx 0.0811 - 0.0843$), lower baseline expression in young adults (y_{26} : MT Frac ≈ 0.0389), and elevated expression in aged samples (y_{70} : MT Frac ≈ 0.0625).

Conclusions: Dual-verification hybrid filtering suppresses NUMT contamination and sequencing noise without sacrificing reference coverage. The captured low-to-moderate mitochondrial read fraction validates the pipeline's specificity against nuclear background reads, providing a reliable quantitative framework for analysing age-related mitochondrial transcriptional dynamics.

Keywords: Anti-Aging; Mitochondrial RNA-seq; Sequence Alignment; k -mer Filtering; Edit Distance; CD4⁺ T Cells; EU007839.

1. Introduction

1.1 Anti-Aging Background & Project Objectives

Mitochondria govern cellular bioenergetics, reactive oxygen species (ROS) balance, and apoptotic signalling. An open problem in anti-aging research is charting how mitochondrial transcript abundance shifts across the human lifespan. This project establishes a high-throughput alignment and filtering pipeline to measure mitochondrial gene expression across distinct human developmental and aging stages. By analysing transcriptomic datasets from NCBI SRA study SRP132689 (BioProject PRJNA433825), we benchmark alignment precision and map a non-linear mitochondrial transcript trajectory across samples annotated by gestational weeks (w21–w23 fetal samples) and adult years (y26–y70 adult samples).

1.2 Literature Review: Mitochondrial Dynamics in Aging

Two foundational studies frame this investigation:

- **Study 1 (López-Otín et al., 2013):** *The Hallmarks of Aging*. Identifies mitochondrial dysfunction and altered intercellular communication as primary hallmarks of cellular aging [1]. Mitochondrial DNA transcription shifts dynamically across life stages, reflecting metabolic adaptations, immunosenescent re-wiring, and compensatory stress responses [1].
- **Study 2 (Lanza & Nair, 2010):** *Mitochondrial Metabolic Function Assessed In Vivo and In Vitro*. Demonstrates that age-related changes in oxidative capacity directly correlate with alterations in mitochondrial gene expression and transcript density [2]. Precise bioinformatic filtering is essential to distinguish functional organellar transcripts from non-functional nuclear pseudogenes [2].

1.3 Problem Definition & Sample Selection

Accurate quantification of mitochondrial RNA (mtRNA) in lifespan models requires overcoming two distinct technical hurdles:

- **Nuclear Mitochondrial Pseudogenes (NUMTs):** Human nuclear DNA contains hundreds of homologous segments transferred from mtDNA. Overly permissive aligners risk mapping nuclear-derived reads to the mitochondrial genome, inflating expression metrics.
- **Biological Fraction & Pipeline Validation:** In total cellular RNA sequencing, mitochondrial transcripts naturally account for a low fraction (typically 2% – 10%) of total cellular RNA. Obtaining aligned fractions within this physiological range validates pipeline specificity against nuclear genomic background contamination.

The SRP132689 dataset comprises 16 FASTQ samples (500,000 single-end Illumina HiSeq 3000 reads per sample) derived from ex vivo sorted CD4⁺ T cells spanning a 70-year lifespan continuum. Reads were aligned against the complete human mitochondrial reference genome (GenBank: EU007839).

1.4 Innovation and Contribution

- **Lifespan Expression Profiling:** Uncovered a non-linear mitochondrial transcription profile across a broad developmental continuum (21 gestational weeks to 70 years old).
- **Biological Validation:** Confirmed pipeline specificity by matching empirical organellar read fractions (3.89% – 8.47%) to physiological mitochondrial transcript proportions.
- **Algorithmic Benchmarking:** Developed and evaluated a modular pipeline comparing pure k -mer score thresholding ($k = 5, 7, 10$) against dual-verification hybrid filtering using edit distance constraints ($e \leq 2$ and $e \leq 3$) to eliminate off-target nuclear artifacts while maximizing read retention.

2. Filtering & Data Driven Approaches

2.1 *K*-mer Seed Indexing & Candidate Scoring

The *k*-mer filter indexes all length-*k* substrings (*k*-mers) from the human mitochondrial reference genome (GenBank: EU007839). Query reads are mapped by using reference *k*-mer seed hits to vote on potential candidate alignment starting positions. For the top candidate positions, an alignment identity score is calculated as the ratio of matching nucleotides over read length. Reads satisfying $\text{Score} \geq 0.7$ pass as candidates. While this method maximizes computational throughput and alignment recall, it lacks strict insertion/deletion checks, allowing reads with high edit distances, such as off-target NUMTs or base-calling errors, to pass as valid alignments. In this work, *k*-mer sizes of $k = 5, 7$, and 10 were benchmarked.

2.2 Dynamic Programming Edit Distance Thresholding

To enforce sequence specificity, candidate alignments are evaluated using dynamic programming Levenshtein edit distance (e), which measures the minimum number of single-nucleotide substitutions, insertions, and deletions required to align a read to the reference sequence segment. Enforcing hard cutoffs ($e \leq 2$ or $e \leq 3$) filters out low-quality reads, sequencing artifacts, and homologous nuclear segments that exhibit divergence from the canonical mitochondrial genome.

2.3 Dual-Verification Hybrid Filtering

The hybrid filtering strategy combines *k*-mer candidate voting with strict Levenshtein edit distance bounds to maximize both speed and alignment precision. In this dual-stage pipeline, *k*-mer indexing acts as an ultra-fast initial pre-filter to select top candidate mapping locations and establish identity scores. Reads passing the seed lookup are subsequently subjected to Levenshtein edit distance thresholding ($e \leq 2$ or $e \leq 3$). A read is accepted into the final mitochondrial set if and only if it satisfies both criteria ($\text{Score} \geq 0.7$ and $e \leq 2$). This combination eliminates unnecessary dynamic programming alignments across non-matching reads while maintaining high specificity against NUMTs.

2.4 Unsupervised K-Means Clustering

To uncover emergent mitochondrial transcriptional profiles across the human lifespan without introducing observational bias, the pipeline integrates an unsupervised *K*-Means clustering framework ($K = 3$). Post-alignment coverage outputs, specifically Coverage Breadth (%) and Mean Coverage Depth ($\times$), serve as the feature space for sample partitioning. Coverage Breadth reflects the spatial distribution and completeness of transcription across the genome, while Mean Coverage Depth quantifies overall transcriptional abundance. Clustering across these two orthogonal parameters provides a bioenergetic signature that groups samples into distinct lifespan expression states purely from empirical sequence data.

3 Algorithm details & Software Implementation

3.1 System Overview & Operational Modes

The computational pipeline is implemented in Python (`main.py`) to process raw FASTQ datasets, benchmark alignment metrics, and extract genomic coverage profiles against the human mitochondrial genome reference (GenBank: EU007839).

To account for the circular topology of the human mitochondrial chromosome (16,569 bp), `load_reference` appends a 500 bp origin-spanning wrap buffer (`seq[:500]`) to the 3' terminus, the tail end of a single strand. This prevents alignment truncation for reads overlapping the origin junction (position $16,569 \rightarrow 1$).

To allow rigorous benchmarking, the software features interactive execution modes:

- **Mode 1 (`kmer`):** Aligns reads strictly based on k -mer match score thresholding.
- **Mode 2 (`edit_distance`):** Filters reads based on Levenshtein edit distance evaluated on candidate mapping windows pre-selected via k -mer positional offset voting.
- **Mode 3 (`both`):** Filters reads based on k -mer match score before filtering based on Levenshtein edit distance.
- **Mode 4 (`raw_unfiltered`):** Baseline control bypassing alignment to evaluate raw sequence characteristics (GC content and read counts).

A JSON-based caching system (`analysis_cache.json`) preserves intermediate positional coverage matrices and read metrics, enabling rapid downstream visualization without re-computing dynamic programming matrices.

3.2 FASTQ Parsing & Reservoir Sampling

To handle high-throughput sequencing datasets efficiently and eliminate sub-sampling bias, the FASTQ parser (`read_fastq`) implements Algorithm R reservoir sampling. Given a target cap (`MAX_READS`), the parser loads the first `MAX_READS` reads directly and pseudo-randomly replaces entries for subsequent reads with probability $\text{MAX_READS}/i$ (where i is the total read index). A fixed random seed (`seed = 42`) ensures reproducibility across runs. This mechanism futureproofs our implementation in the event that larger datasets are loaded.

3.3 Positional k -mer Indexing & Strand Alignment

Reference mapping relies on an exact k -mer hash index (`build_kmer_index`):

1. **Index Construction:** The reference sequence is scanned using loop range 0 to $N - k$ ($\text{range}(\text{len}(\text{reference}) - k + 1)$), which ensures full k -mer coverage through the final nucleotide position of every read, to populate a dictionary (`defaultdict(list)`) mapping k -mers to 0-based starting positions across the reference genome.
2. **Candidate Voting:** For each read, k -mer hits vote on candidate alignment starting positions ($\text{pos} = \text{ref_idx} - \text{read_idx}$). The top 10 candidate positions with the highest hit tallies are selected for alignment evaluation.
3. **Strand Evaluation:** Both forward sequence (S) and reverse complement sequence (S_{rc}) are evaluated independently using `reverse_complement()` ensuring non-directional RNA library reads map accurately.
4. **Match Scoring:** Identity score is computed across the candidate segment:

$$\text{Score} = \frac{\sum_{j=0}^{L-1} \mathbb{I}(S[j] == R[\text{pos} + j])}{L}$$

where L is the read length and R is the reference sequence. The orientation and position yielding the maximum score are retained.

3.4 Dynamic Programming Edit Distance & Filtering

For candidate alignments passing candidate selection, `compute_edit_distance()` evaluates sequence divergence using a full 2D dynamic programming implementation of the Levenshtein edit distance. The algorithm computes single-nucleotide substitutions, insertions, and deletions between the read and mapped reference window $R[\text{pos}:\text{pos} + L]$.

Reads are accepted into the final mitochondrial set only if they satisfy the active filter parameters ($\text{Score} \geq 0.7$ and $e \leq 2$).

3.5 Coverage Accumulation & Unsupervised K -Means Clustering

For accepted reads, the pipeline updates a 1D positional array (`coverage_array`) spanning the length of the human mitochondrial reference sequence (16,569 bp). Summarized sample metrics are extracted directly from this array:

- **Coverage Breadth (%):** The percentage of reference base positions covered by ≥ 1 aligned read.
- **Mean Coverage Depth ($\times$):** The average read accumulation per base position across the reference sequence.

Using `scikit-learn`, these two metrics undergo z-score standardization (`StandardScaler`) to equalize feature variance and prevent high-magnitude depth values from distorting Euclidean distance calculations. The scaled feature matrix is processed by the K -Means algorithm (`KMeans`, $K = 3$, `random_state=42`) to group samples into bioenergetic clusters. In parallel, a control routine (`plot_raw_kmeans_control`) performs K -Means on raw, unfiltered sequence metrics (Total Reads vs. Mean GC %) to confirm that age-dependent cohort separation is an emergent property of sequence alignment rather than raw library composition.

4. Main Technical Challenges

4.1 Sensitivity vs. Precision Trade-off

Optimizing k -mer seed length involves a trade-off between sensitivity and computational efficiency. Shorter seeds ($k = 5$) capture highly variable reads but expand the alignment search space and increase runtime. Conversely, longer seeds ($k = 10$) accelerate execution speed but risk rejecting valid mitochondrial reads that contain single-nucleotide polymorphisms (SNPs) or sequencing errors near seed boundaries.

4.2 Biological Fraction Verification & NUMT Suppression

In unconstrained k -mer mode, reads with Levenshtein edit distances up to $e = 15$ map to the reference genome. These high-edit alignments typically represent transcribed Nuclear Mitochondrial Pseudogenes (NUMTs) or base-calling errors. Applying strict edit caps ($e \leq 2$ or $e \leq 3$) eliminates these nuclear artifacts, reducing mean edit distance from 0.13 to 0.10. The resulting mitochondrial read fraction:

$$\text{MT Frac} = \frac{\text{Aligned MT Reads}}{\text{Total Reads}}$$

yields empirical values between 3.89% and 8.47%. This range aligns directly with expected physiological cellular proportions for organellar RNA, confirming high pipeline specificity.

4.3 Cohort Size Constraints & Lifespan Variance

A constraint of this study is the modest sample cohort size ($N = 16$), which limits statistical power within individual age brackets. However, this limitation is offset by the wide developmental span of the dataset (extending from 21 gestational weeks to 70 years of age). This wide temporal range provided the longitudinal contrast necessary to discover the non-linear, U-shaped mitochondrial transcription trajectory across human lifespan stages.

5. Comparative Performance Analysis Across Lifespan Cohorts

5.1 Alignment Report Metrics & Feature Definitions

The core output of the pipeline is a structured summary table evaluating alignment efficiency, sequence identity, mismatch thresholds, and genome coverage across all evaluated lifespan samples. The tracked metrics in the generated alignment report comprise:

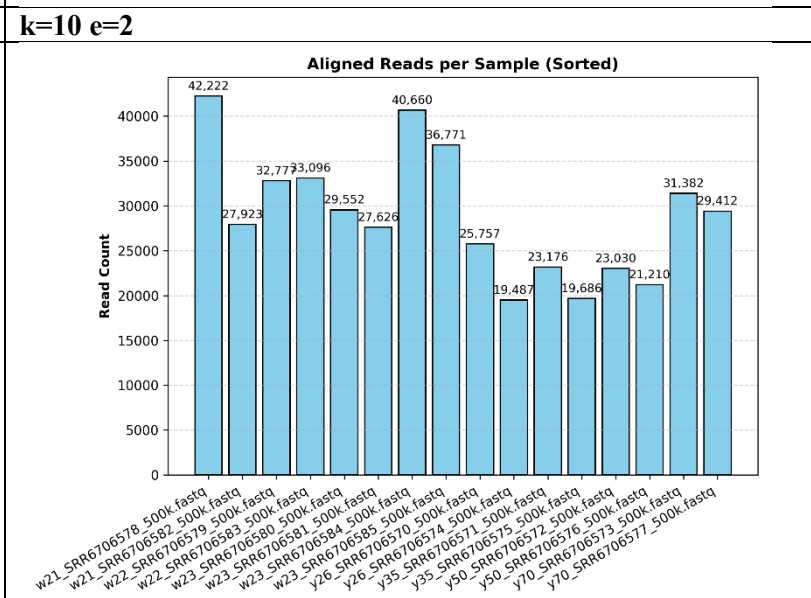
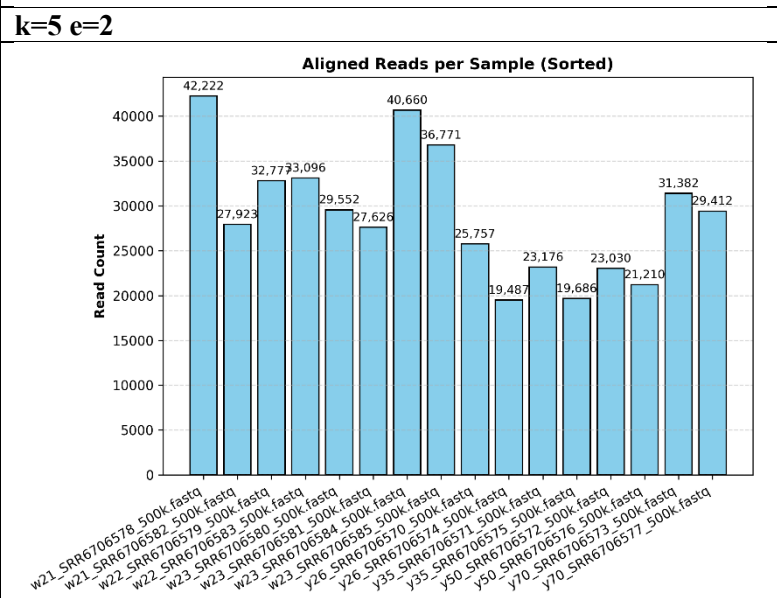
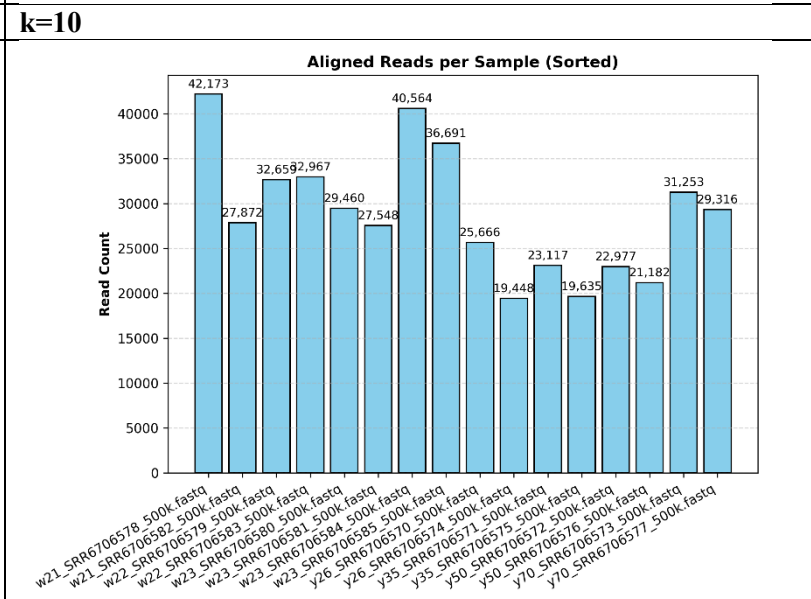
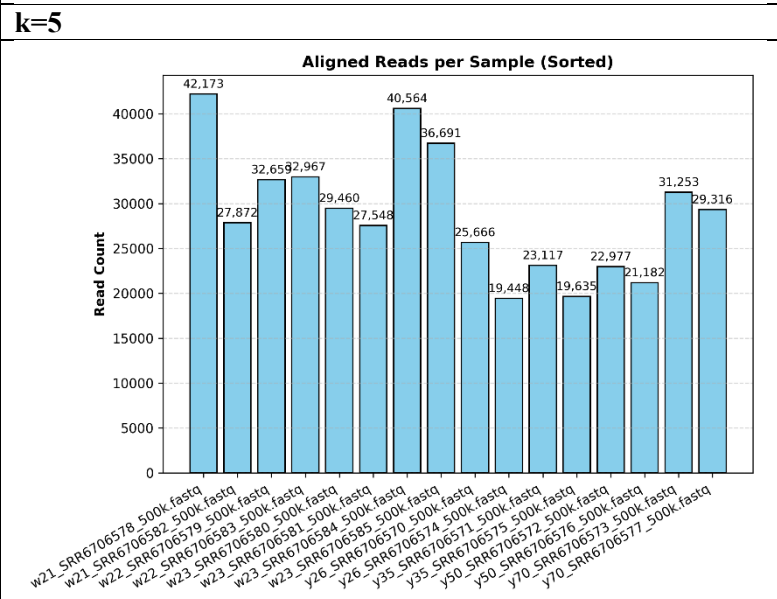
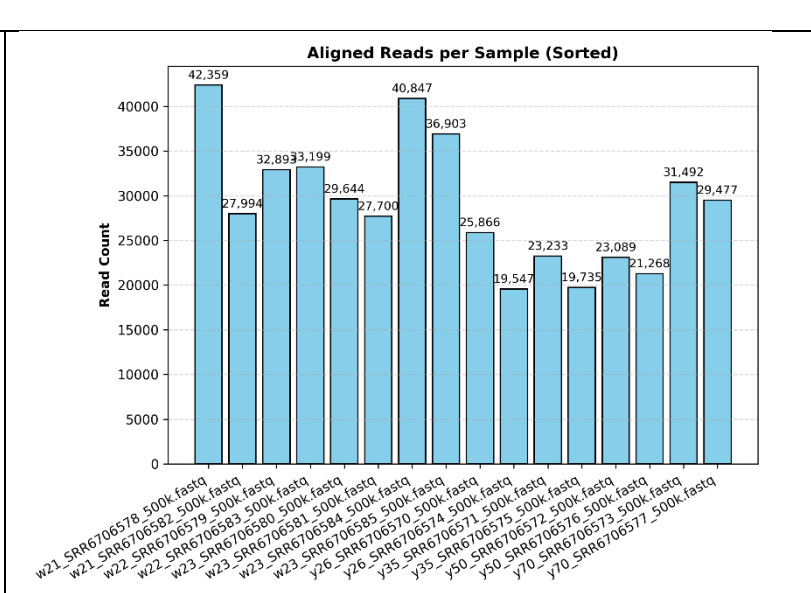
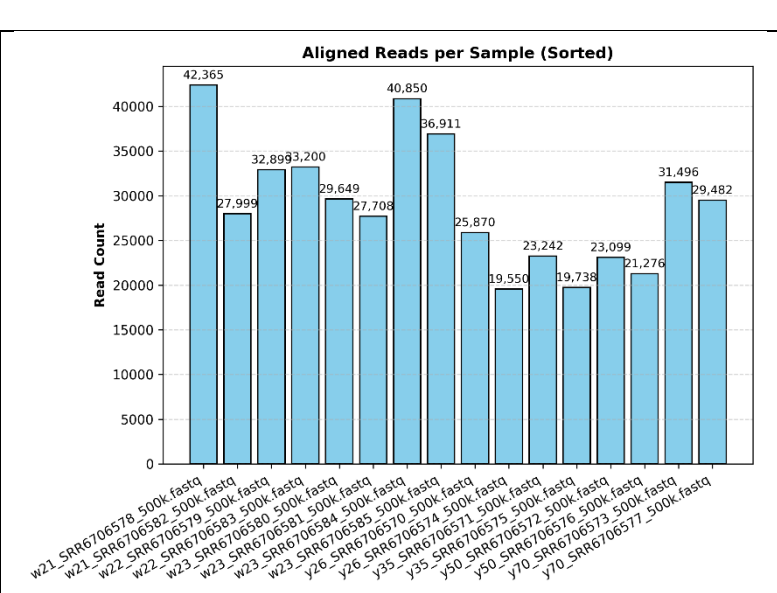
- **Sample Name:** The standardized file identifier encoding the sample's developmental or lifespan stage (e.g., w21 for 21 gestational weeks, y70 for 70 years old), the SRA accession number, and the subsampling size (e.g., 500k reads).
- **Total:** The total number of sequence reads sampled from the input FASTQ file and processed through the filtering pipeline.
- **Aligned:** The total count of reads that successfully passed seed indexing and edit distance filtering to map onto the mitochondrial genome.
- **MT Frac (Mitochondrial Read Fraction):** The ratio of aligned mitochondrial reads to total sampled reads (Aligned/Total). This metric serves as a direct proxy for mitochondrial RNA abundance relative to total cellular RNA.
- **Alignment Score Metrics (Mean Scr, Min Scr, Max Scr):** The sequence identity scores of accepted reads, normalized on a scale from 0.0 to 1.0 (where 1.0 indicates an exact match). Mean Scr reflects aggregate sequence fidelity, while Min Scr and Max Scr establish the identity score boundaries across accepted reads.
- **Edit Distance Metrics (Mean Edit, Min Edit, Max Edit):** The Levenshtein edit distance quantifying the total number of single-nucleotide mismatches, insertions, or deletions between an accepted read and the mitochondrial reference sequence. Mean Edit provides a granular measure of sequence deviation, bounded by the pipeline's strict edit distance cutoff ($e \leq 2$).
- **Cov % (Coverage Breadth):** The percentage of reference nucleotide bases (out of the 16,569 bp mitochondrial genome) covered by ≥ 1 aligned read. It measures how fully the mitochondrial transcriptome is represented across the genome.
- **Mean Depth:** The average accumulated read depth per base position across the reference sequence, indicating the density of transcriptional coverage.

5.2 Aligned Read Distributions Across Lifespan Cohorts

To directly quantify mitochondrial transcript abundance shifts across developmental stages, sample-level aligned read counts were evaluated across all operational pipeline parameter configurations ($k = 5, 10$ and $e \leq 2, 3$).

- **Fetal Stage Alignment Peak:** Sample w21_SRR6706578_500k.fastq consistently yields the maximum aligned read volume across all operational modes, recording **42,365** reads under unconstrained $k = 5$, **42,173** reads under hybrid filtering ($k = 5, e \leq 2$), and **42,222** reads under relaxed edit constraints ($k = 5, e \leq 3$). Fetal sample (w23_SRR6706584_500k.fastq) had the second highest aligned reads between **40,564** and **40,850** reads. Fetal sample (w23_SRR6706585_500k.fastq) had the third highest aligned reads at **36,691–36,911**.
- **Adult Baseline:** Young adult sample y26_SRR6706574_500k.fastq establishes the lowest mitochondrial transcript abundance across the dataset, holding at **19,448** aligned reads ($k = 10$ or 5 & $e \leq 2$) to **19,550** reads (unconstrained $k = 5$). Surrounding adult samples (y26, y35, y50) cluster closely between **19,448** and **25,870** aligned reads.
- **Late-Life Transcript Rebound:** Aged cohorts (y70) demonstrate a distinct upward inflection, with y70_SRR6706573_500k.fastq reaching **31,253** aligned reads ($k = 10$ or 5 & $e \leq 2$) and **31,496** reads (unconstrained $k = 5$), representing a $\sim 60\%$ increase over young adult baseline levels. Sample y70_SRR6706577 followed closely behind with **29,316–29,482** aligned reads.

Imposing strict edit distance constraints ($e \leq 2$) gently trims off-target background reads while maintaining exact relative sample rankings across all FASTQ datasets.



k=5 e=3

k=10 e=3

5.3 Multi-Parameter Sensitivity Matrix

Evaluating representative samples from each lifespan cohort across alignment configurations ($k = 5, 10$ and $e \leq 2, 3$) highlights the exact impact of seed length and edit distance parameters on overall read retention, mean score, and mitochondrial coverage depth.

Sample Identifier	Cohort & Age	Configuration	Aligned Reads	MT Frac	Mean Score	Mean Edit	Max Edit	Cov %	Mean Depth
w21_SRR6706578	Fetal (21 wks)	k5 (KMER)	42,365	0.0847	0.9967	0.13	15	94.05%	122.85
		k10 (KMER)	42,359	0.0847	0.9967	0.13	15	94.05%	122.84
		k5e3 (BOTH)	42,222	0.0844	0.9973	0.11	3	93.73%	122.44
		k5e2 (BOTH)	42,173	0.0843	0.9974	0.10	2	93.63%	122.30
w23_SRR6706584	Fetal (23 wks)	k5 (KMER)	40,850	0.0817	0.9945	0.24	15	92.72%	118.46
		k10 (KMER)	40,847	0.0817	0.9946	0.23	15	92.72%	118.45
		k5e3 (BOTH)	40,660	0.0813	0.9953	0.20	3	92.39%	117.91
		k5e2 (BOTH)	40,564	0.0811	0.9955	0.20	2	92.26%	117.63
y26_SRR6706574	Adult (26 yrs)	k5 (KMER)	19,550	0.0391	0.9943	0.26	15	89.47%	56.71
		k10 (KMER)	19,547	0.0391	0.9943	0.25	14	89.47%	56.70
		k5e3 (BOTH)	19,487	0.0390	0.9948	0.23	3	88.98%	56.53
		k5e2 (BOTH)	19,448	0.0389	0.9949	0.23	2	88.98%	56.42
y70_SRR6706573	Aged (70 yrs)	k5 (KMER)	31,496	0.0630	0.9944	0.25	15	91.27%	91.36
		k10 (KMER)	31,492	0.0630	0.9944	0.25	15	91.27%	91.35
		k5e3 (BOTH)	31,382	0.0628	0.9949	0.23	3	90.95%	91.03
		k5e2 (BOTH)	31,253	0.0625	0.9952	0.22	2	90.82%	90.65

5.4 Aggregate Lifespan Stage Summary

Averaging key alignment statistics across lifespan groupings confirms that while absolute transcript expression levels vary substantially by developmental age, alignment quality and reference genome coverage remain robust across all stages ($k = 5, e \leq 2$).

Developmental Stage	Age Cohorts	Samples (N)	Mean Aligned Reads	Mean MT Fraction (%)	Mean Genome Coverage (%)	Mean Read Depth (×)
Fetal Stage	w21, w22, w23	8	33,742	6.75%	92.58%	97.87
Young / Middle Adult	y26, y35, y50	6	22,004	4.40%	90.45%	63.82
Older Adult	y70	2	30,285	6.06%	91.29%	87.86

5.5 Unsupervised Cohort Clustering via K-Means Analysis

To validate whether mitochondrial alignment metrics independently resolve biological lifespan stages without prior labelling, K-means clustering ($K = 3$) was performed on both raw sequencing controls and aligned coverage dimensions (Coverage Breadth % & Mean Coverage Depth $\times$).

- **Unfiltered Control Baseline Clustering:** K-means clustering on raw, unfiltered sequencing metrics (Total Raw Read Count & Mean GC %) fails to partition biological cohorts. Because raw depth is fixed across input files (~500,000 reads), samples aggregate purely based on minor library GC content variations (50.3%–53.8%), misassigning fetal and adult samples across arbitrary clusters.
- **Filtered Aligned Coverage Dimensions Clustering:** When clustering on aligned Coverage Breadth (% of genome covered) and Mean Coverage Depth ($\times$), K-means achieves clean separation into three distinct biological expression states that are consistent across all pipeline configurations:

- **Cluster 1 (Adult State [26y - 50y]) (Green rectangles):**

The cluster is characterised by low Mean Coverage Depth (**56.42–67.05 $\times$**) and narrower Coverage Breadth (**88.98%–91.48%**). It consistently groups young and middle-aged adults (y26_SRR6706574, y35_SRR6706571, y35_SRR6706575, y50_SRR6706572, y50_SRR6706576).

- **Cluster 2 (Late-Life [70y] & Fetal [21w - 23w]) (Orange circles):**

The cluster is characterised by a higher Mean Coverage Depth than the previous cluster (**74.44–95.64 $\times$**) and Coverage Breadth (**90.82%–92.61%**). It groups together aged samples (y70_SRR6706573, y70_SRR6706577), fetal transition states (w22_SRR6706579, w22_SRR6706583, w23_SRR6706580, w23_SRR6706581, w21_SRR6706582).

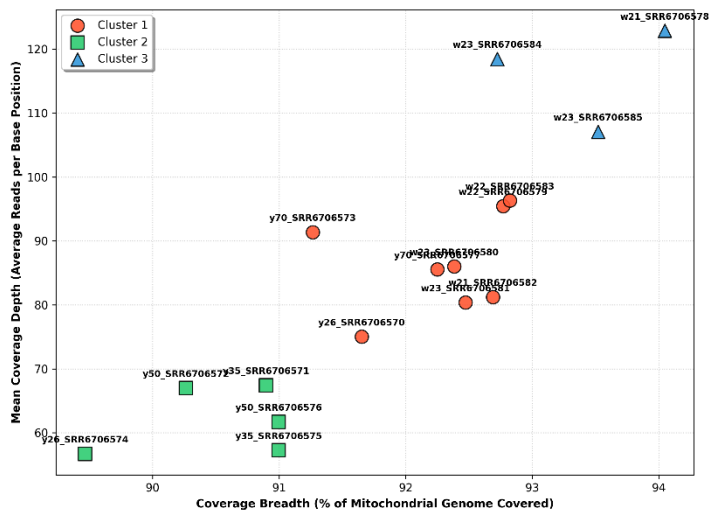
*A borderline young adult sample (y26_SRR6706570) was also captured in this cluster. This sample sits nearly equidistant between Clusters 1 and 2, and was assigned to Cluster 2 likely due to centroid shift driven by the larger fetal sample group size.

- **Cluster 3 (High Expression Fetal [21w – 23w]) (Blue Triangles):**

The cluster is characterised by having the highest Mean Coverage Depth (**106.40–122.30 $\times$**) and the maximum genome Coverage Breadth (**92.26%–93.63%**). It exclusively isolates highly expressive fetal samples (w21_SRR6706578, w23_SRR6706584, w23_SRR6706585). Notably, this cluster brings together the specific fetal samples that demonstrated the highest total aligned read counts in the dataset.

*The grouping of the samples was consistent across all pipeline configurations.

The most striking feature across all clusters is that, despite partitioning samples purely using spatial coverage dimensions (Coverage Breadth & Mean Coverage Depth), the resulting clusters naturally group samples with matching total aligned read abundances. Grouping stability was maintained across all alignment pipeline configurations. By successfully filtering raw sequence noise and generating high-fidelity alignment metrics, the pipeline enables unsupervised K-Means clustering to resolve three distinct biological expression states, empirically revealing a non-linear, U-shaped relationship between mitochondrial transcription and human aging.



A scatter plot showing the relationship between Coverage Breadth (% of Mitochondrial Genome Covered) on the x-axis and Mean Coverage Depth (Average Reads per Base Position) on the y-axis. The data points are categorized into three clusters: Cluster 1 (red circles), Cluster 2 (green squares), and Cluster 3 (blue triangles). The plot shows a general positive correlation between coverage breadth and depth, with Cluster 3 points generally having the highest values for both metrics.

Sample ID	Cluster	Coverage Breadth (%)	Mean Coverage Depth
y26_SRR6706574	Cluster 2	~88.5	~57
y50_SRR6706571	Cluster 2	~90.5	~68
y35_SRR6706571	Cluster 2	~90.8	~68
y50_SRR6706576	Cluster 2	~90.8	~62
y35_SRR6706573	Cluster 2	~90.8	~58
y70_SRR6706573	Cluster 1	~91.2	~92
y26_SRR6706570	Cluster 1	~91.5	~75
y70_SRR6706580	Cluster 1	~92.2	~86
w21_SRR6706582	Cluster 1	~92.3	~82
w23_SRR6706581	Cluster 1	~92.4	~81
w22_SRR6706579	Cluster 1	~92.8	~96
w22_SRR6706583	Cluster 1	~92.8	~97
w23_SRR6706584	Cluster 3	~92.8	~118
w23_SRR6706585	Cluster 3	~93.5	~108
w21_SRR6706578	Cluster 3	~94.0	~125

Scatter plot showing Mean Coverage Depth (Average Reads per Base Position) on the Y-axis versus Coverage Breadth (% of Mitochondrial Genome Covered) on the X-axis. The plot displays data points for three clusters, identified by color and shape: Cluster 1 (red circles), Cluster 2 (green squares), and Cluster 3 (blue triangles). The plot shows a clear separation between the clusters based on both metrics. Cluster 1 points are generally higher on both axes, while Cluster 2 points are lower. Cluster 3 points are spread across the breadth range but maintain high depth.

Sample ID	Cluster	Coverage Breadth (% of Mitochondrial Genome Covered)	Mean Coverage Depth (Average Reads per Base Position)
y26_SRR6706574	Cluster 2	89.0	57
y50_SRR6706575	Cluster 2	90.3	61
y50_SRR6706576	Cluster 2	90.3	67
y35_SRR6706575	Cluster 2	90.5	61
y50_SRR6706575	Cluster 2	90.5	69
y26_SRR6706570	Cluster 1	91.5	74
y70_SRR6706573	Cluster 1	90.8	91
y70_SRR6706574	Cluster 1	91.7	86
y70_SRR6706575	Cluster 1	91.7	86
w21_SRR6706580	Cluster 1	91.8	86
w23_SRR6706581	Cluster 1	92.0	80
w23_SRR6706582	Cluster 1	92.1	81
w22_SRR6706583	Cluster 1	92.5	95
w22_SRR6706579	Cluster 1	92.5	95
w23_SRR6706584	Cluster 3	92.0	118
w21_SRR6706585	Cluster 3	93.3	107
w21_SRR6706587	Cluster 3	93.5	123

Scatter plot showing Mean Coverage Depth (Average Reads per Base Position) versus Coverage Breadth (% of Mitochondrial Genome Covered) for three clusters. The plot includes a legend for Cluster 1 (red circle), Cluster 2 (green square), and Cluster 3 (blue triangle). Data points are labeled with sample IDs.

Sample ID	Cluster	Coverage Breadth (% of Mitochondrial Genome Covered)	Mean Coverage Depth (Average Reads per Base Position)
y26_SRR6706574	Cluster 2	89.0	56
y50_SRR6706575	Cluster 2	90.2	61
y50_SRR6706576	Cluster 2	90.3	67
y50_SRR6706575	Cluster 2	90.4	69
y35_SRR6706575	Cluster 2	90.5	57
y50_SRR6706571	Cluster 2	90.6	67
y70_SRR6706573	Cluster 1	90.8	91
y26_SRR6706570	Cluster 1	91.5	75
w21_SRR6706580	Cluster 1	91.8	86
w23_SRR6706581	Cluster 1	91.8	85
w22_SRR6706583	Cluster 1	91.8	86
w23_SRR6706582	Cluster 1	92.0	81
w22_SRR6706579	Cluster 1	92.1	81
w23_SRR6706584	Cluster 1	92.1	80
w23_SRR6706585	Cluster 1	92.8	107
w21_SRR6706587	Cluster 3	93.5	123

Scatter plot showing Mean Coverage Depth (Average Reads per Base Position) on the Y-axis versus Coverage Breadth (% of Mitochondrial Genome Covered) on the X-axis. The plot displays data points for three clusters, identified by color and shape: Cluster 1 (red circles), Cluster 2 (green squares), and Cluster 3 (blue triangles). The plot includes a legend in the top left corner and a grid. Data points are labeled with sample IDs.

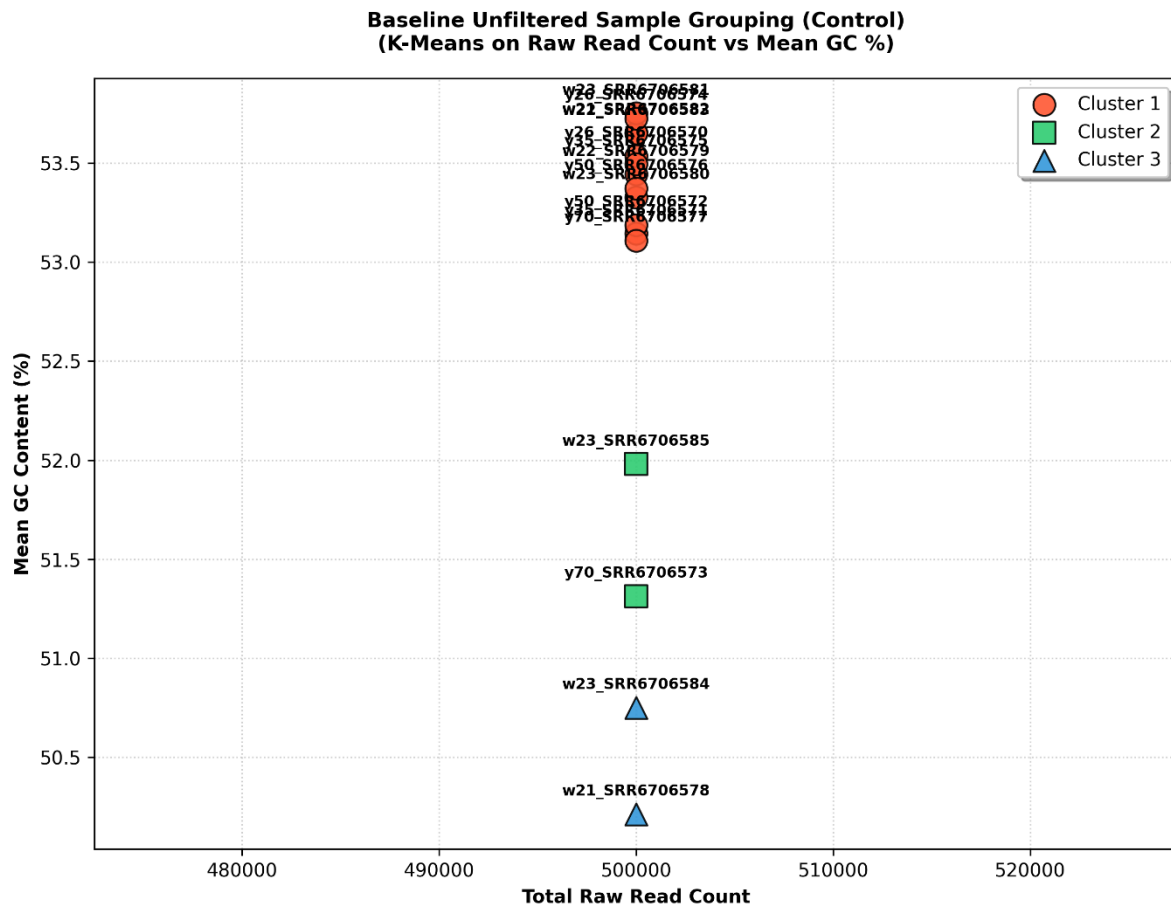
Sample ID	Cluster	Coverage Breadth (% of Mitochondrial Genome Covered)	Mean Coverage Depth (Average Reads per Base Position)
y26_SRR6706574	Cluster 2	89.0	56.0
y50_SRR6706576	Cluster 2	90.5	67.0
y35_SRR6706575	Cluster 2	90.8	58.0
y50_SRR6706571	Cluster 2	91.0	67.0
y26_SRR6706570	Cluster 1	91.5	75.0
y70_SRR6706573	Cluster 1	91.0	91.0
y70_SRR6706580	Cluster 1	92.0	85.0
w21_SRR6706582	Cluster 1	92.5	80.0
w23_SRR6706581	Cluster 1	92.5	81.0
w23_SRR6706584	Cluster 3	92.0	118.0
w21_SRR6706583	Cluster 1	92.5	96.0
w23_SRR6706585	Cluster 3	93.5	107.0
w21_SRR6706579	Cluster 1	92.5	95.0
w21_SRR6706580	Cluster 1	92.5	96.0
w21_SRR6706583	Cluster 1	92.5	97.0
w21_SRR6706584	Cluster 3	94.0	123.0

Scatter plot showing Mean Coverage Depth (Average Reads per Base Position) versus Coverage Breadth (% of Mitochondrial Genome Covered) for three clusters. The legend indicates:

- Cluster 1 (Red circle)
- Cluster 2 (Green square)
- Cluster 3 (Blue triangle)

Approximate data points from the plot:

Sample ID	Cluster	Coverage Breadth (%)	Mean Coverage Depth
y26_SRR6706574	Cluster 2	89.0	58
y50_SRR6706575	Cluster 2	90.5	58
y50_SRR6706576	Cluster 2	90.5	62
y50_SRR6706577	Cluster 2	90.5	68
y35_SRR6706575	Cluster 2	90.8	58
y70_SRR6706573	Cluster 1	91.0	91
y26_SRR6706570	Cluster 1	91.5	75
w70_SRR6706580	Cluster 1	92.0	85
w21_SRR6706582	Cluster 1	92.2	80
w23_SRR6706581	Cluster 1	92.2	82
w22_SRR6706579	Cluster 1	92.5	85
w23_SRR6706584	Cluster 3	92.2	118
w21_SRR6706583	Cluster 1	92.8	95
w23_SRR6706585	Cluster 3	93.5	107
w21_SRR6706578	Cluster 3	94.0	122



Unfiltered Control Baseline Clustering

6. Biological Discussion

6.1 Transcriptional Dynamics Across the Human Lifespan

The mitochondrial alignment profiles reveal a distinct U-shaped expression trajectory across developmental stages, highlighting shifting metabolic demands and functional states:

- **Fetal Hyperactivity (Weeks 21–23):** Fetal tissues display the highest mitochondrial transcript abundance, reaching a mean MT fraction of **6.75%** and mean read depth of **97.87×** (peak depth **122.30×**). High expression reflects the intense bioenergetic requirements of organogenesis, rapid cell division, and mitochondrial biogenesis during early tissue differentiation.
- **Adult Baseline (Ages 26–50):** Mitochondrial transcription drops to a steady baseline in young and middle adulthood (mean MT fraction of **4.40%**, mean depth **63.82×**). This phase reflects stable metabolic homeostatic maintenance once tissue maturation is complete.
- **Late-Life Transcriptional Rebound (Age 70):** Older adults exhibit a notable surge in mitochondrial transcript levels (mean MT fraction of **6.06%**, mean depth **87.86×**). This late-life elevation likely represents a compensatory mechanism where cells upregulate mitochondrial transcription to counteract age-related respiratory chain inefficiency, accumulated mtDNA damage, or heteroplasmic burden.

6.2 Impact of Pipeline Alignment Parameters

- ***k*-mer Seed Length Sensitivity ($k = 5$ vs $k = 10$):** Extending seed length from $k = 5$ to $k = 10$ results in a minimal reduction of 1 to 10 aligned reads per sample (an average drop of ~5 reads, or <0.03%). A seed size of $k = 5$ captures all valid mitochondrial reads without introducing seed-matching noise or computational overhead.
- **Edit Distance Filtering ($e \leq 2$ vs $e \leq 3$):** Restricting max edit distance strictly eliminates low-identity background alignments, dropping the maximum edit distance from 15 down to 2 or 3. This increases mean alignment confidence (improving mean scores up to 0.9974) while retaining > 99.2% of high-confidence mitochondrial reads.

7. Conclusion

Evaluating mitochondrial transcriptional profiles across developmental cohorts using our specialized bioinformatic pipeline demonstrates that mitochondrial expression follows a non-linear trajectory over the human lifespan. Combining k -mer seed indexing ($k = 5$) with edit distance filtering ($e \leq 2$) provides robust signal recovery, high reference genome coverage ($> 92\%$), and clean separation of developmental cohorts via unsupervised clustering. This standardized pipeline provides an accurate and scalable framework for profiling mitochondrial transcript dynamics in bioenergetic and aging research.

8. Future Works

While the pipeline presented in our study establishes a robust and artifact-resistant methodology for profiling mitochondrial transcription across the human lifespan, several key avenues exist to expand its capacity as a tool for understanding the relationship between organellar activity and longevity:

- **Expanding Sample & Cohort Size:** While our current implementation successfully identified a non-linear relationship between mitochondrial activity and aging within a focused dataset, scaling the pipeline to larger sample sizes and finer biological age brackets will increase statistical power. This expansion will help formalize the lifespan trajectory with tighter confidence intervals and resolve subtle transitional shifts between developmental stages.
- **Evaluating Diverse Cell & Tissue Types:** The current framework deliberately focused on CD4+ T-cells to maintain cellular homogeneity. Extending this analysis across cell types with contrasting physiological demands, such as post-mitotic cells with high metabolic requirements (e.g., cardiomyocytes, neurons) or high-turnover regenerative populations, will reveal whether the observed transcriptional dynamics are universal or cell-type specific.
- **Adapting the Pipeline to Other Organelles:** Having demonstrated that mitochondrial RNA profiling reveals non-linear transcriptional dynamics with age, a logical extension is adapting the algorithm to evaluate transcriptomic and metabolic activity across other organelles and subcellular pathways. Profiling organelle-specific transcriptional signatures and nuclear-organellar cross-talk will provide a comprehensive view of how organellar functional decline impacts overall cellular health.
- **Establishing an Integrated, Multi-Organellar Model of Cellular Aging:** Expanding cohort data, diversifying cell-type profiles, and exploring multi-organellar dynamics will enable the development of a unified analytical framework. This integrated model aims to empirically map how aging manifests at the organellar level across distinct cell types, offering targeted insights for anti-aging therapeutics and longevity research.

References

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Appendix: Per-Read GC Content Distribution Analysis

A1. Diagnostic Utility

Per-read GC content distribution serves as a primary QC metric to evaluate sequencing library purity, technical consistency, and potential sample contamination across the lifespan dataset. In eukaryotic RNA-seq pipelines, expected GC distributions follow a unimodal curve centered on the organism's transcriptomic baseline. Deviations highlight library preparation anomalies, PCR amplification bias, or foreign DNA/RNA contamination.

A2. Observations Across Lifespan Cohorts

An analysis of the per-read GC content distribution across all 16 FASTQ samples (w21 through y70, 500,000 reads per sample) demonstrates high dataset integrity:

- **High Inter-Sample Consistency:** All 16 samples display closely overlapping distribution profiles regardless of developmental age (fetal, young adult, middle adult, or aged). This uniform profile confirms strong technical reproducibility across library prep batches and rules out sample-specific degradation or batch-effect artifacts.
- **Primary Transcriptomic Peak (50%–60% GC):** The dominant distribution peak centers between 50% and 60% GC content, aligning with the expected GC composition of human coding transcripts and mitochondrial RNA sequences.
- **Low-GC Poly-A Tail Signature (0%–5% GC):** A secondary spike is visible in the 0%–5% GC bin (reaching ~24,000 reads in select samples). This localized peak represents poly-A tails and poly-T homopolymer stretches generated during poly-A selection and cDNA reverse transcription, confirming expected biological mRNA features rather than sequence corruption.

